

Cops and Robbers: A Graph Theory Game

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Outline

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Motivation

- Imagine a city with a single cop actively pursuing a robber.
- Both know the city layout and each other's location.
- They take alternating turns moving to adjacent locations.
- **Question:** Can the robber escape safely and outrun the cop?
- We model this using a connected, simple, finite graph $G(V, E)$.

The Game of Cops and Robbers

The Game

Let $G(V, E)$ be a connected, simple, finite graph. The rules are:

- 1 **Starting Positions:** Cop chooses $c_0 \in V(G)$, then robber chooses $r_0 \in V(G)$.
- 2 **Movement:** Players move alternately, cop goes first. A player at x can move to adjacent y (where $xy \in E(G)$) or stay at x .
- 3 **Perfect Information:** Both players know the graph structure and each other's position at all times.

Cop-win and Robber-win Graphs

Cop-win Graph

A graph G is *cop-win* if there is a winning strategy for the cop, meaning the cop guarantees $c_t = r_t$ for some finite turn t .

Robber-win Graph

A graph G is *robber-win* if there is a winning strategy for the robber, meaning the robber guarantees $c_t \neq r_t$ for all turns $t \geq 0$.

Lemma

All complete graphs K_n are cop-win graphs.

Proof Idea: Cop starts anywhere. Robber picks a vertex. Cop catches robber on the first move since all vertices are adjacent.

Lemma

All trees are cop-win graphs.

Proof Idea: There is a unique path between any two vertices in a tree. The cop simply takes the unique path towards the robber, strictly decreasing the distance at each step.

Cycles are Robber-win

Lemma

All cycles of length at least 4 are robber-win graphs.

Proof Idea:

- Robber starts at distance ≥ 2 from the cop (possible since $n \geq 4$).
- Every vertex has exactly two neighbors.
- The robber simply moves in the opposite direction of the cop.
- The cop can never decrease the distance to 0, avoiding capture indefinitely.

Pitfalls

To formalize a general characterization, we define a *pitfall*.

Pitfall

A vertex $u \in V$ is a **pitfall** (or corner) if there exists a distinct vertex $v \in N(u)$ such that its closed neighborhood is a subset of the closed neighborhood of v , i.e., $N[u] \subseteq N[v]$. We say v **dominates** u .

Intuition

If the robber is at a pitfall u , and the cop is at the dominating vertex v , the cop can reach any vertex the robber could move to, guaranteeing capture.

The Dismantlability Theorem

Theorem

*A finite graph is cop-win **iff** it can be reduced to a single vertex through the successive removal of pitfalls.*

Forward Direction ($\Rightarrow$):

- Consider the turn before capture. Cop at v must be able to reach any vertex robber at u can move to.
- This implies $N[u] \subseteq N[v]$, meaning u is a pitfall.
- Removing u preserves the strategy, so the graph is dismantlable.

The Dismantlability Theorem (cont.)

Backward Direction ($\Leftarrow$):

- Proceed by induction by removing pitfall u dominated by v to get restricted graph G' .
- Construct a mapping $f(x)$ mapping u to v .
- The cop chases the "shadow" robber on the smaller cop-win graph G' .
- Once caught in G' , if the real robber is at u , they are immediately caught from v as $N[u] \subseteq N[v]$.

The Cop Number

Cop Number

The cop number of a graph G , denoted $\mathcal{C}(G)$, is the minimum number of cops required to guarantee catching the robber.

- Note that $\mathcal{C}(G) \leq \beta(G)$, where $\beta(G)$ is the domination number of G .
- If the cops form a dominating set, they can capture the robber in one step.

Isometric Path Lemma

Lemma (Isometric Path Lemma)

Let P be a shortest path between vertices $u, v \in V(G)$. With at least two cops in play, one cop on P can prevent the robber from ever entering P . (If the robber steps on P , they are immediately caught).

Proof Sketch:

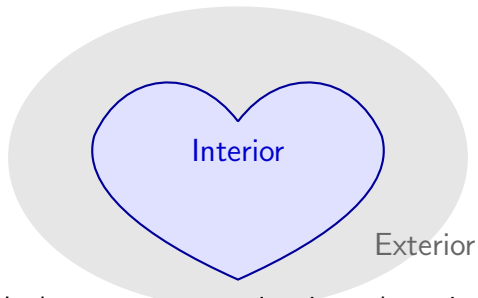
- Cop maintains strategy ensuring $d(c, z) \leq d(r, z)$ for all $z \in V(P)$.
- If the robber threatens $z_0 \in V(P)$ by decreasing distance, the cop moves towards z_0 .
- Robber cannot threaten two vertices x, y on opposite sides of the cop simultaneously, as P is a shortest path (violates triangle inequality).
- Thus, if the robber steps on $z \in V(P)$, $d(c, z) \leq d(r, z) = 0$, meaning the cop captures them.

Theorem (Aigner and Fromme)

Any planar graph G has $\mathcal{C}(G) \leq 3$.

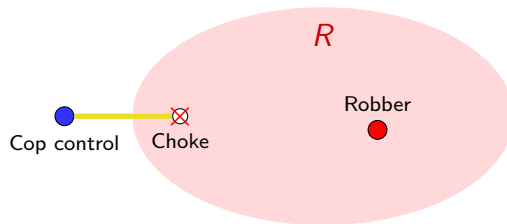
- *Idea:* Use 2 cops to control two shortest paths meeting at a vertex, forming a territorial boundary.
- Use the Jordan Curve Theorem on these paths and the 3rd cop on a crossing path to strictly reduce the robber's bounded territory at each stage.
- Eventually, the territory is strictly reduced until the robber is caught.

Jordan Curve Illustration

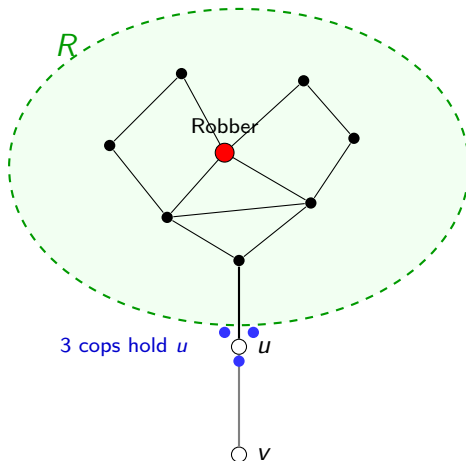


Jordan curve separates interior and exterior

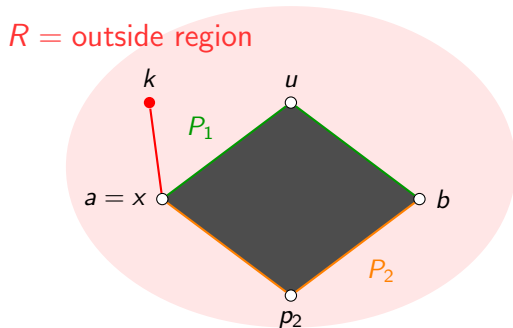
Robber Territory: Reachable Zone



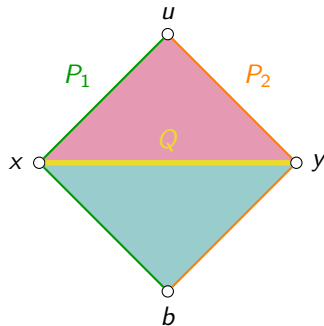
Case A: One Bottleneck Toward the Territory



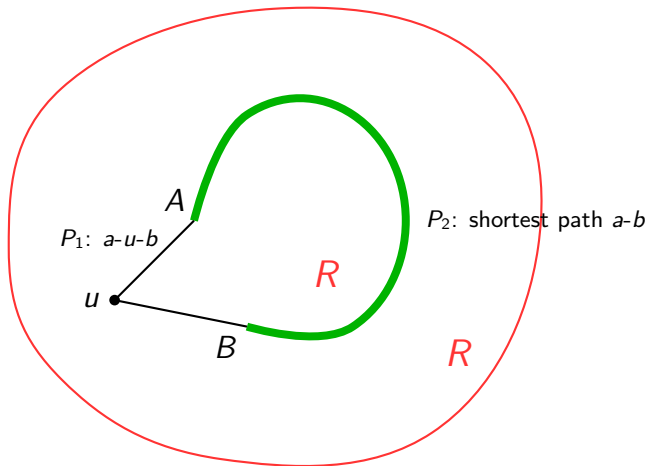
Case B(i): Cycle with Disjoint External Component



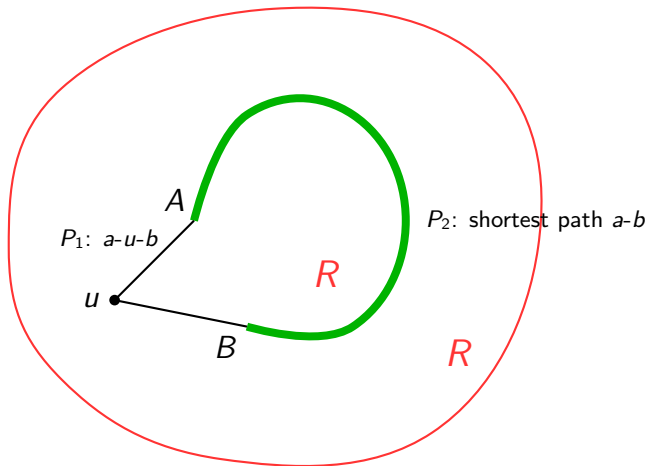
Case B(ii): Path Q Splitting the Cycle



Case B(ii) Details: Path Q Starting from A



Case B(ii) Details: Path Q Starting from u



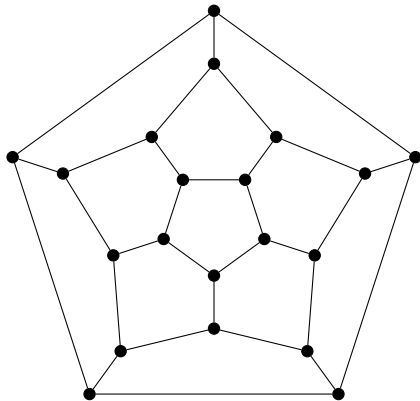
A Lower Bound Theorem

Theorem

Any graph G with no 3-cycles or 4-cycles and $\delta(G) \geq n$ has $\mathcal{C}(G) \geq n$.

- *Proof Sketch:* Place $n - 1$ cops.
- The robber can find a vertex r not dominated by any of the $n - 1$ cops.
- Due to the absence of 3- and 4-cycles, after cops move, at most $n - 1$ of the robber's neighbors are threatened.
- Since the robber has at least n neighbors ($\text{min degree} \geq n$), there is always at least one safe neighbor to continually move to!

Tightness of Planar Bound



This planar graph has no 3- or 4-cycles, and $\delta(G) = 3$. By our previous theorem, $\mathcal{C}(G) \geq 3$, proving the planar bound is tight.

- The game of Cops and Robbers intimately connects pursuit-evasion dynamics with structural graph theory.
- **Single Cop:** Cop-win graphs are exactly those that are dismantlable via pitfalls.
- **Multiple Cops:** The minimum degree in graphs with large girth limits the cop number from below.
- **Planarity:** Topologically, planar graphs drastically restrict evasion, capping the cop number at 3.

Thank You!